

Exam II Review Topics

1. Exponential Functions 2.4:31-34(31,32)
 - a. Properties
Graph
Domain, Range, Asymptote, Monotonicity
 - b. Algebraic Properties
 - c. Solving Exponential Equations
2. Logarithm Functions == Inverses of Exponential Functions 2.4:23-26,39-40(39,40)
 - a. Properties
Graph
Domain, Range, Asymptote, Monotonicity
 - b. Algebraic Properties
 - c. Solving Logarithmic Equations
 - d. Natural Logarithm
Properties
3. Exponential Growth/Decay 2.4:66
4. Derivative of $y = f(x)$ at a point $x = a$ 3.1:13-16,20-23(22,23)
 - a. Slope of the secant line to the graph of $y = f(x)$ over $[a, a + h]$
 - b. Slope of the tangent line to the graph of $y = f(x)$ at $x = a$
 - c. Average rate of change of $y = f(x)$ over $[a, a + h]$
 - d. Instantaneous rate of change of $y = f(x)$ at $x = a$
 - e. Difference quotient for $y = f(x)$ over $[a, a + h]$
 - f. Definition of derivative of $y = f(x)$ at $x = a$
5. Equation of the tangent line to $y = f(x)$ at a point $x = a$ 3.1:29-30; 3.2:27-31(28,30);
3.3:45-50(45,49); 3.5:41-45(41,45);
3.6:33-36(34)
 - a. Point-slope formula
6. Relationship between the graphs of $y = f(x)$ and $y' = f'(x)$ 3.1:6-9
7. (Non-)Existence of a derivative for a function $y = f(x)$ at a point $x = a$

- a. Function f undefined at $x = a$
 - b. Function f discontinuous at $x = a$
 - c. Graph of $y = f(x)$ has a corner at $x = a$
- 8. Differentiability of $y = f(x)$ at $x = a$ implies continuity of $y = f(x)$ at $x = a$
- 9. Rules for differentiation 3.2:11-20(13,20)
 - a. Constant rule
 - b. Power rule
 - c. Sum rule
 - d. Difference rule
 - e. Product rule
 - f. Quotient rule
- 7. Derivatives
 - a. Polynomials
 - b. Rational Functions
 - c. Algebraic Functions
- 8. Higher Order Derivatives 3.2:21-26(26); 3.3:33-41(41); 3.6:45-46(46)
- 9. Rules of Special Functions 3.3:9-26(10,17,26)
 - a. Rules for special functions
 - a. Sine function rule
 - b. Cosine function rule
 - c. Other trig function rules (via quotient rule)
 - d. Exponential function (e^x)
 - e. (Natural) logarithm function ($\ln x$)
- 10. Rectilinear Motion 3.4:17-20,36-37(18,36)
 - position function $s = s(t)$
 - a. velocity function $v = s'(t)$
 - acceleration function $a = v'(t) = s''(t)$
 - b. Direction of motion determined by v
 - c. Total distance travelled
- 11. Projectile Motion 3.4:41-45(41,44)
 - a. $g \approx 32 \text{ ft/sec}^2 \approx 9.8 \text{ m/sec}^2$
 - b. $h(t) = s_0 + v_0 t - \frac{1}{2} g t^2$
 - c. $v = 0$ at maximum height
- 12. Chain rule 3.5:13-35(19,25,26,30)

- a. Extended derivative rules
13. Implicit Differentiation 3.6:1-12(5,6,12)
- a. General procedure
 - i. Equation $F(x, y) = 0$
 - ii. Treat (locally) y as a (unknown, but specified) function $f(x)$
 - iii. Differentiate the equation (both sides) w.r.t. x (writing y' for the derivative of y (the unknown, but specified function $f(x)$))
 - iv. Solve (linear) equation for y'
 - b. Tangent line construction
 - c. Higher order derivatives
14. Differentiation rules for special functions 3.6:19-25(20,21)
- a. Roots ($y = \sqrt{x} = x^{1/2}$, $y = \sqrt[n]{x} = x^{1/n}$)
 - b. Powers ($y = x^r$)
 - c. Exponentials ($y = b^x$)
 - d. Logarithms ($y = \log_b x$)
 - e. Inverse trig functions
 - i. $y = \sin^{-1} x$
 - ii. $y = \tan^{-1} x$
 - iii. Others ($y = \sec^{-1} x$, $y = \cos^{-1} x$)
15. Logarithmic Differentiation 3.6:50-55(51,54)
- a. Products, Quotients, Powers
 - b. Variable Base - Exponential Functions
16. Related Rate Problems 3.7:14-15,23-30(24,26)
- a. Procedure (Page 158)